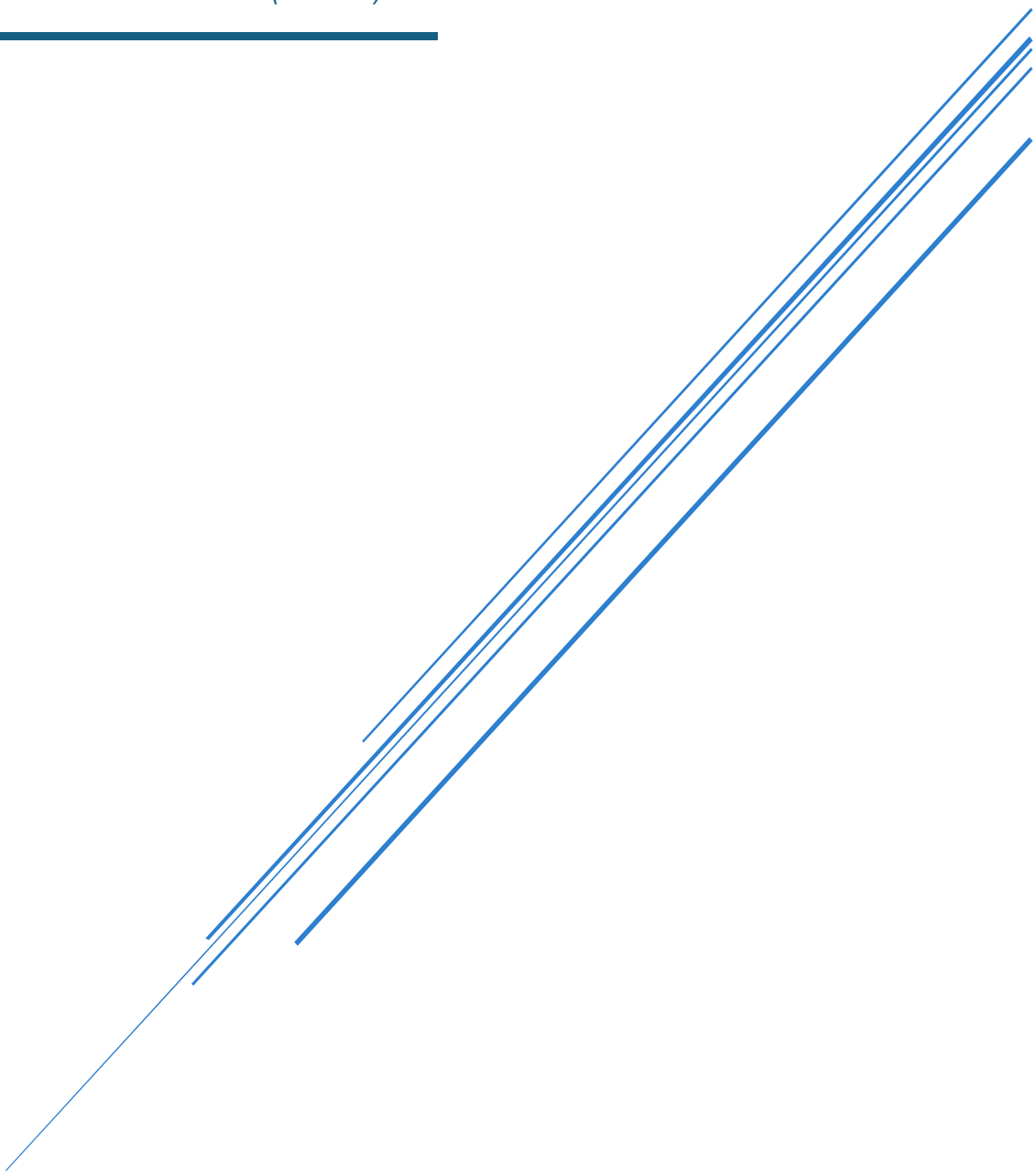
PRG181 – GROUP 3

Project 2

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Section 1: Problem Solving Logic: Library Book Management

1. Introduction

A library book management system is a program that controls the recording of book information in a library. In addition to maintaining accurate data for the books in the library, this program will have the ability to look up specific books based upon their title or authors' name. This system also provides the means for a patron to check out and return the books they borrow. A menu driven interface allows users to easily navigate through the different features offered.

2. Problem Definition

The library book management system must be able to meet the following needs:

- store & maintain records of books
- permit users to look up books based upon their title or authors' name
- allow users to check out & return books
- display all books that are currently available
- prevent multiple patrons from checking out the same book
- Handle any invalid input entered by the user correctly
- continue operating until the patron has completed all tasks

3. System Decomposition

In designing the logic of this system, we broke down its design into small manageable pieces:

- add book
- search book
- check out book
- return book
- show list of books available for checkout
- Exit the program

By using this modular approach, we are improving our systems' modularity, readability and maintainability.

4. Control logic (menu driven design)

We used a simple menu driven control logic where there is a loop allowing users to pick one of several choices while the system is operational; then once the user picks "Exit", the program stops.

Logic flow:

- start the system
- display options to the user
- collects the option selected by the user
- executes the selected operation
- returns to the main menu
- continues looping until the user selects the "Exit" option

This approach helps improve usability and ensures that your program will run consistently.

5. Data Structures Design

5.1 Book Representation

Each book is stored as an object with the following characteristics:

- book id
- title
- authors name
- status (available/unavailable)

5.2 Data Storage

-The system stores and manages each book's data in a predetermined list of books.

6. Input-Process-Output (IPO) Model

6.1 Inputs

- the system utilizes a predefined book collection (list)
- user selection from the menu
- details about a particular book (id, title, author)
- criteria to use when performing a search (Title or Author)
- id number of the book being borrowed or returned

6.2 Processes

Add Book

- Validate the entered input data
- Verify that the id number entered does not already exist for another book
- create a new record representing the added book
- set the status of the new book record to "available"
- store the new record to the collection

Search Book

- accept search parameters entered by the patron
- iterate over the collection of books
- compare the search parameters entered by the patron to those stored in each book's record
- display all matching records

Borrow Book

- prompt user for id number of book they want to check-out
- find the record representing the desired book
- determine if the status property is equal to "available"
- if status = "available", change the status property to "borrowed"; otherwise, display an error message

Return Book

- prompt patron for id number of book being returned
- locate the book with the specified id number
- determine if the status is equal to "borrowed"
- Update status to "available" if valid; otherwise, display an error message

Display Available Books

- traverse through collection of books
- only display those records which have status = "available"

Exit command

- Terminate program

6.3 Outputs

- Confirmation messages to the patron (e.g., success)
- Error messages to the patron (e.g., book not located)
- List of books to the patron (e.g., search results, available books)

Section 2: Pseudocode

Main Program

```
START PROGRAM
DECLARE books AS LIST
Declare sNewID, sNewTittle ,sNewAuthor, sSearch, sBorrow, sReturn, sSearchResult : = ""
Declare iAmount , iCOunt : = 0
Declare bRunning as Boolean, bFound as Boolean
SET bRunning = TRUE
WHILE running = TRUE
        DISPLAY "========================"
        DISPLAY " LIBRARY MENU "
        DISPLAY "========================"
        DISPLAY "1. Add Book"
        DISPLAY "2. Search Book"
        DISPLAY "3. Borrow Book"
        DISPLAY "4. Return Book"
        DISPLAY "5. Display All Books"
        DISPLAY "6. Exit"
        DISPLAY "========================"
        INPUT =get value
        CASE choice OF
         1: CALL AddBook()
         2: CALL SearchBook()
         3: CALL BorrowBook()
         4: CALL ReturnBook()
         5: CALL DisplayBooks()
         6: SET bRunning = FALSE ; DISPLAY "Program terminated" ;
         DEFAULT: DISPLAY "Invalid choice. Enter a number between 1 and 6."
        END CASE
END WHILE
END PROGRAM
```

Add Book Procedure

```
Begin
DISPLAY "========================"
DISPLAY " ADD BOOK "
DISPLAY "========================"

Set sNewID = Get value(Prompt "Please Enter Book ID")


FOR (iCOunt=0, icount < Length(book)) DO
        IF book[0] = sNewID THEN
         DISPLAY "Error: That ID already exists."
         RETURN
        END IF
END FOR

Set sNewTitle = Get value (Prompt "Enter book Tittle ")
Set sNewAuthor = Get value (Prompt "Enter Author Name")
SET newBook = [newID, newTitle, newAuthor, "Available"]
ADD newBook TO book
DISPLAY "Book added successfully!"
END PROCEDURE
```

Search Procedure

```
Begin
DISPLAY "========================"
DISPLAY " SEARCH BOOK "
DISPLAY "========================"
Set sSearch =Upcase(Get Value (Prompt "Search by tittle or author"))
SET iAmount = 0
SET iCount = 0
SET sSearchResult = ""
While  iCount >= (length(book)-1) DO
       If (sSearch == book[iCount][2]) or (sSearch == book[iCount][3]) // Search by author and tittle
        Set sSearchResult += #11 + book[iCount][1]+",  by " +book[iCount][2] + " : " + book[iCount][3]
        iAmount ++
       END IF
       iCount ++
END WHILE

IF iAmount != 0 THEN SET sSearchResult = "Error no results found"
Display sSearchResult

END PROCEDURE
```

Borrow Book Procedure

```
Begin
DISPLAY "========================"
DISPLAY " BORROW BOOK "
DISPLAY "========================"
SET sBorrow = Upcase(Get Value(Prompt "Enter bookID or Tittle to borrow"))
SET bFound = False
SET iCount = 0
WHILE (iCount >= (length(book) -1) or (bFound = false) DO
        IF sBorrow == Book[iCount][2] or sBorrow ==Book[iCount][1] THEN
                IF Book[iCount][3] = "Available" THEN
                        SET Book[iCount][3] = "Unavailable"
                        SET bFound = True
                END IF
                Else Then
                        Display"Sorry "+Book[iCount][1]+" is already borrowed"
                        Return
                End Else If
        END IF
        Else iCount ++
END WHILE
IF bFound = FALSE THEN
DISPLAY "No book found with ID: " + borrowID
END IF
Else Display "Succesfully borrowed book"
END PROCEDURE
```

Return Book Procedure

```
Begin
DISPLAY "========================"
DISPLAY " RETURN BOOK "
DISPLAY "========================"
Set sReturn = Upcase(Get Value (Prompt "Which book do you want to return"))
SET iCount = 0
SET bFound = False
While ( iCount <= (length(Book) -1) ) or (bFound = false) do
       IF sReturn ==  Book[iCount][0] or sReturn == Book[iCount][1] THEN
              SET bFound = TRUE
              IF Book[iCount][3] == "Availiable" THEN
                     DISPLAY book[i][1] + " has been returned successfully."

                     ELSE
                     DISPLAY book[i][1] + " was not borrowed."
              END IF
       END IF
END While
IF bFound = FALSE THEN
Display "Error book not found"
END IF
END PROCEDURE
```

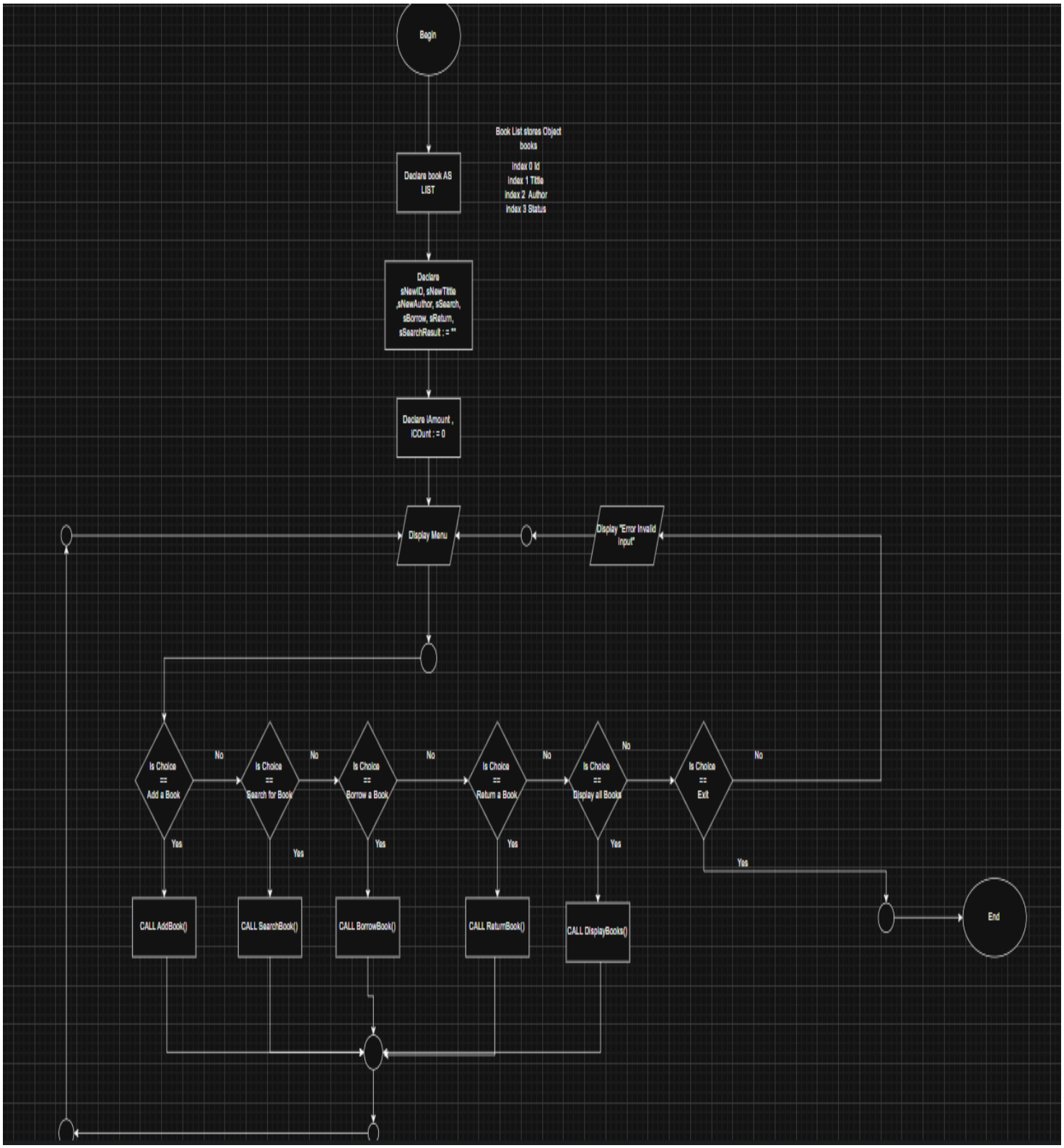
Display All Books Procedure

```
Begin
DISPLAY "========================"
DISPLAY " ALL BOOKS "
DISPLAY "========================"
IF book IS EMPTY THEN
 Set sDisplay = "List not yet populated"
 RETURN
END IF
FOR i = 0 TO LENGTH(books) - 1 DO
       Set sDisplay =Book[iCount][0] + ": "+Book[iCount][1] +"by"+Book[iCount[2]+ #9 + Book[iCount][3]
END FOR

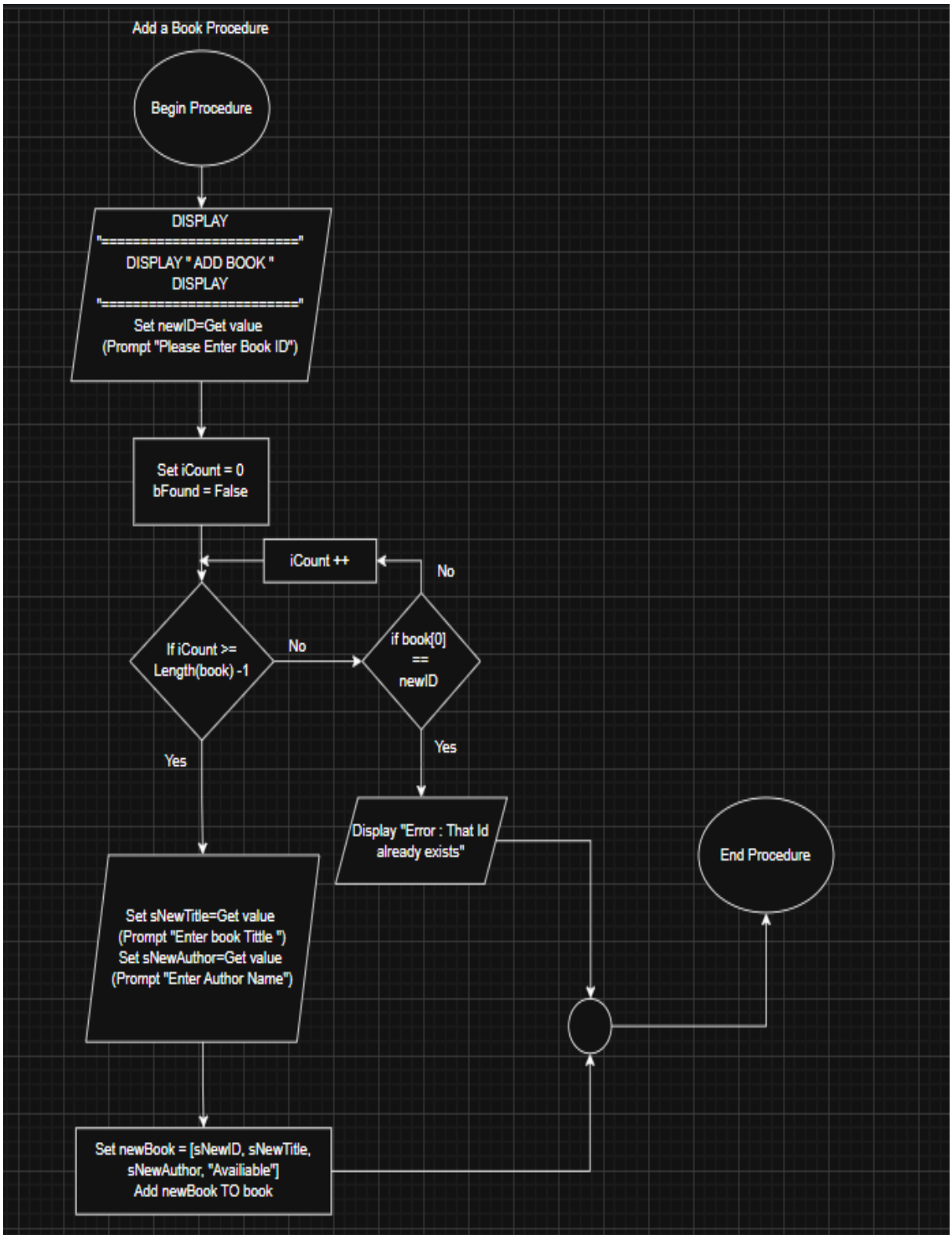
Display sDisplay
END PROCEDURE
```

Section 3: Flowchart

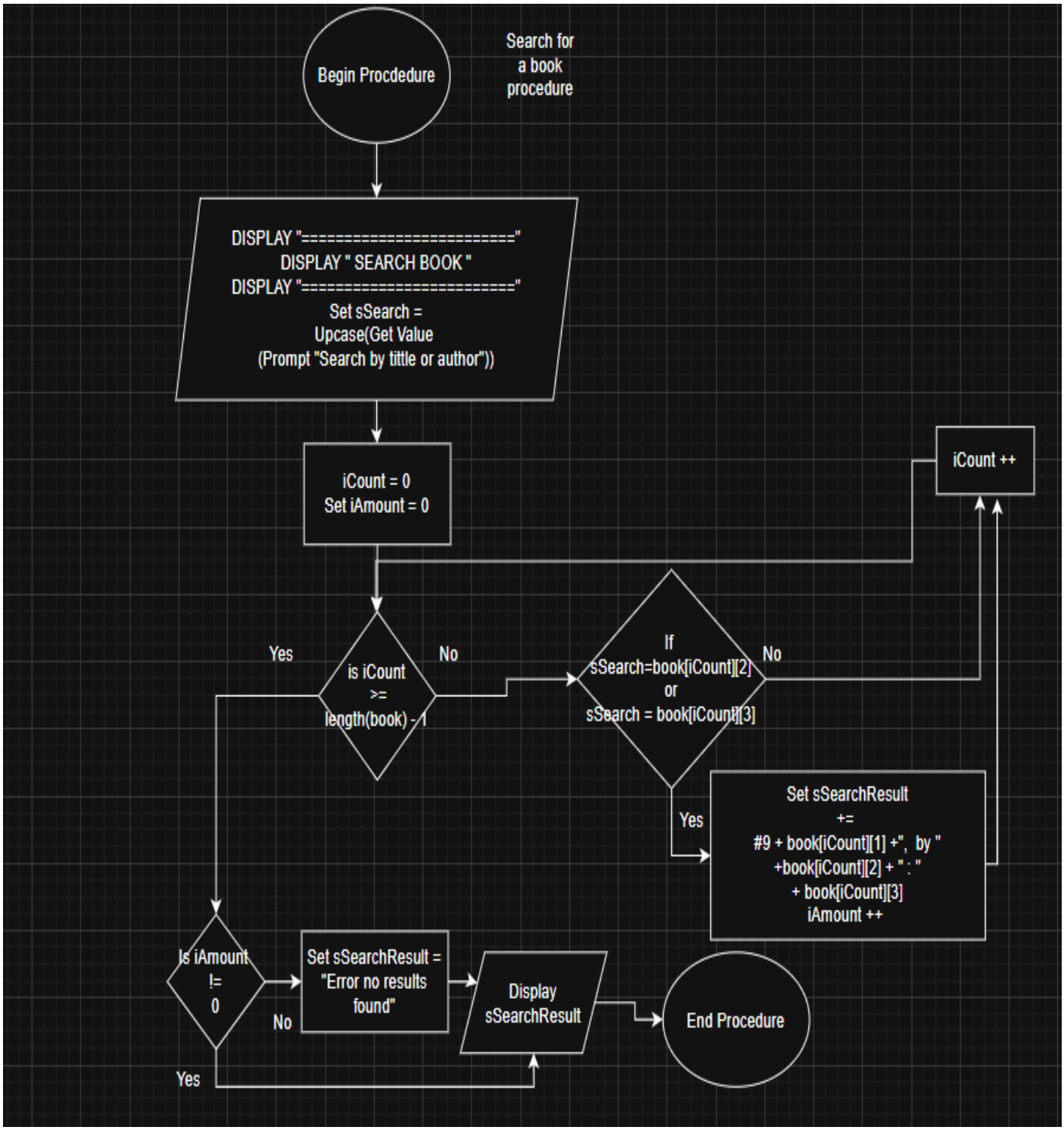
Main Program



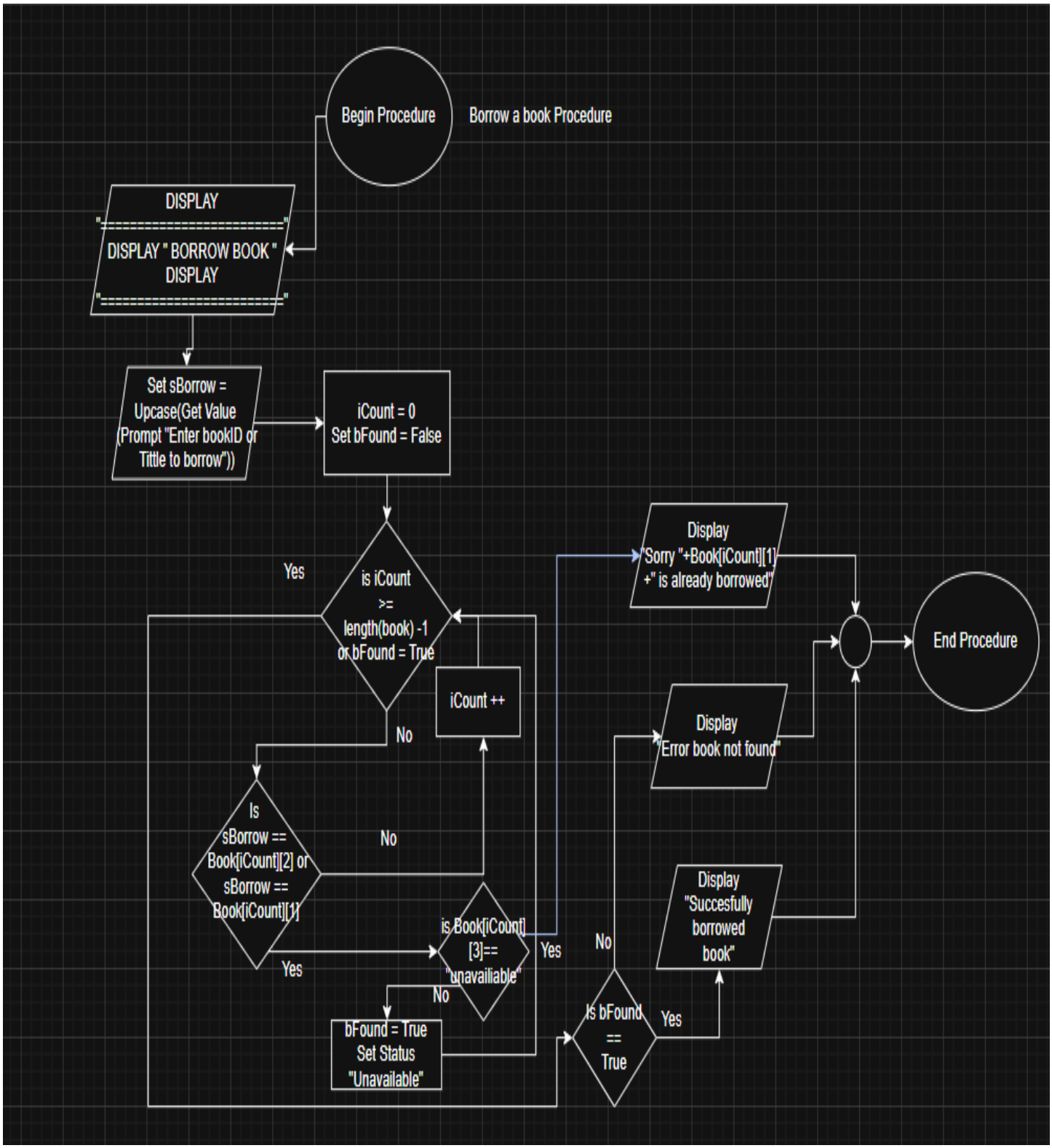
Add Book Procedure



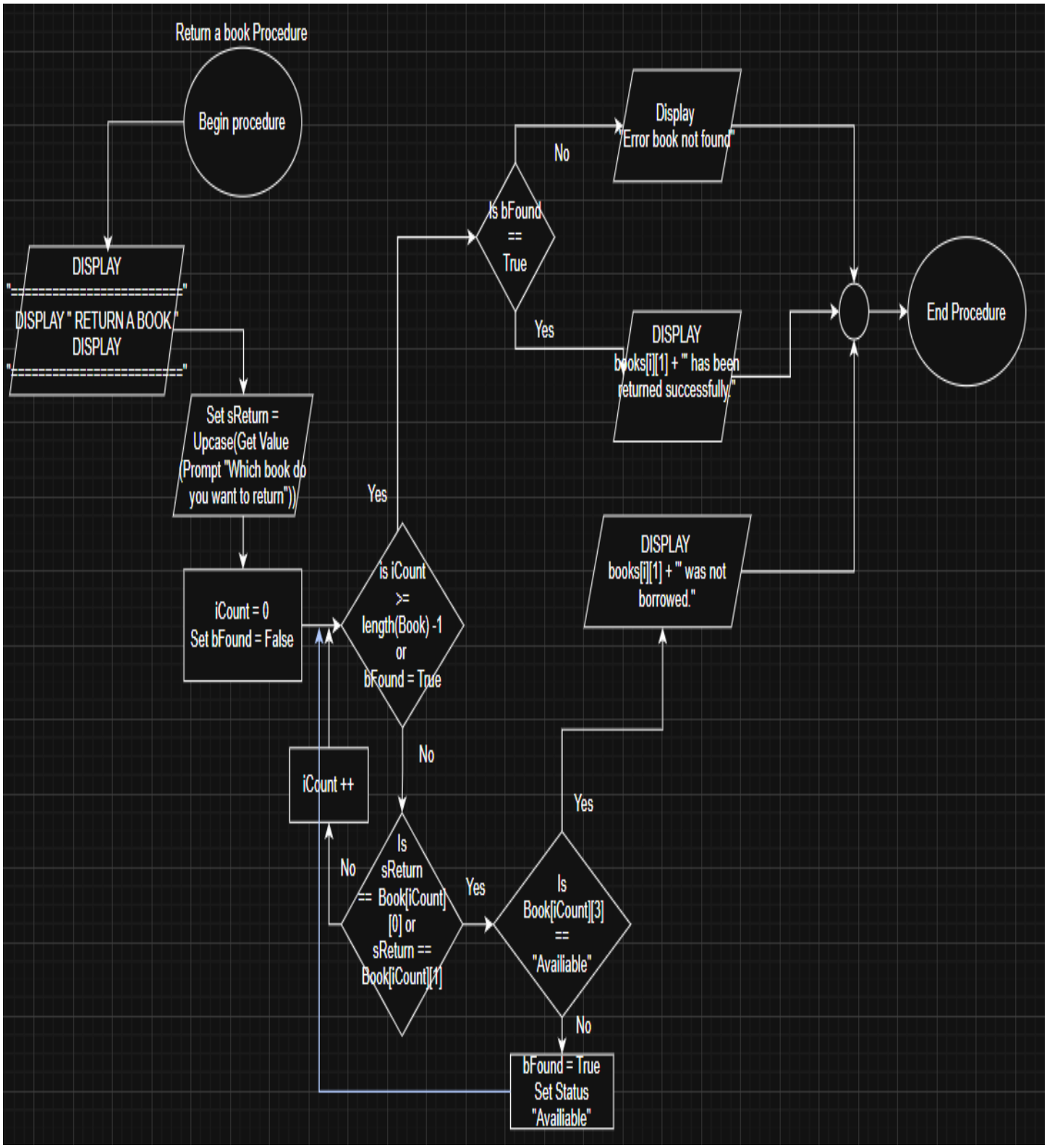
Search Procedure



Borrow Book Procedure



Return Book Procedure



Display All Books Procedure

